CMNIST – Data & Abduction Diagnostics
Dataset: cmnist16_c00_clean
Source folder: data/cmnist16_c00_clean

This report summarizes:

- Residual diagnostics & counterfactual consistency (U-Net)
- HL Image_ distribution across interventions
- Background & edge analysis for U_ll_hat
- Gaussianity tests (LL & HL noise)

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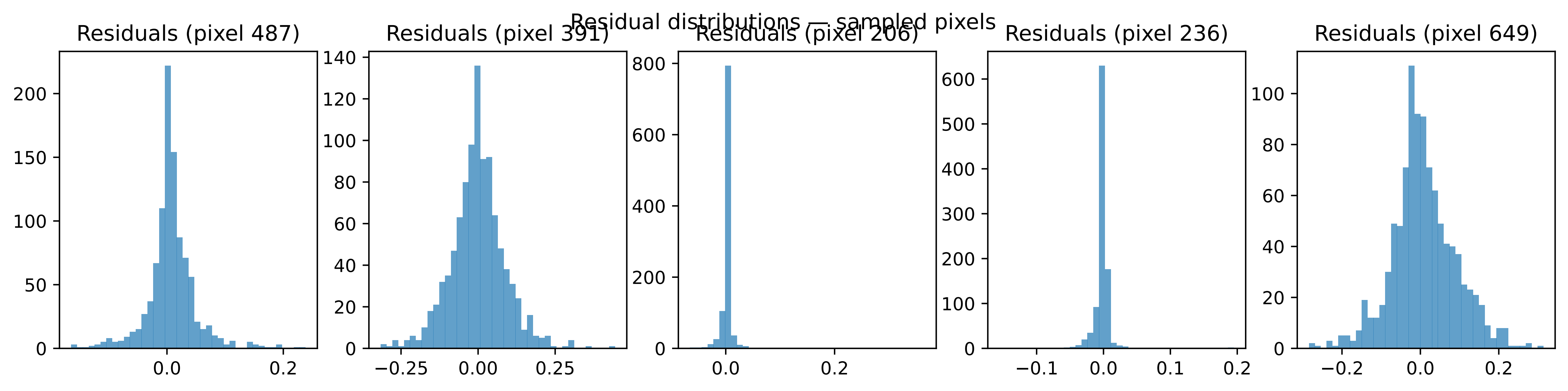
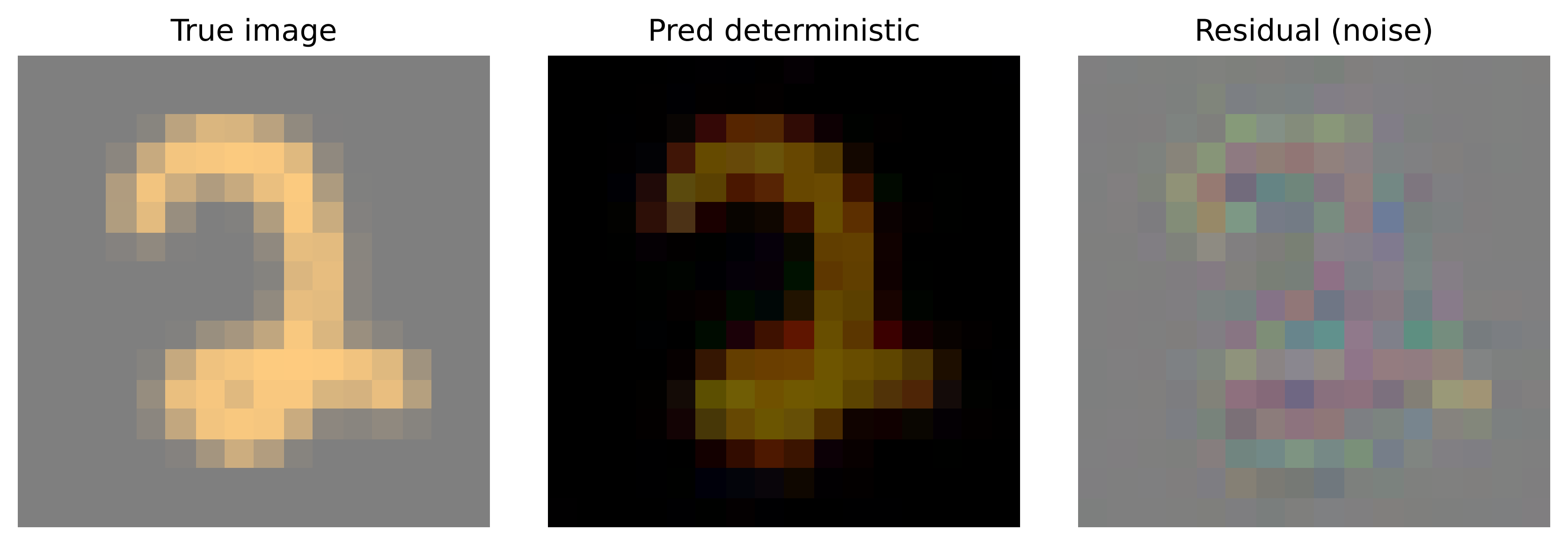
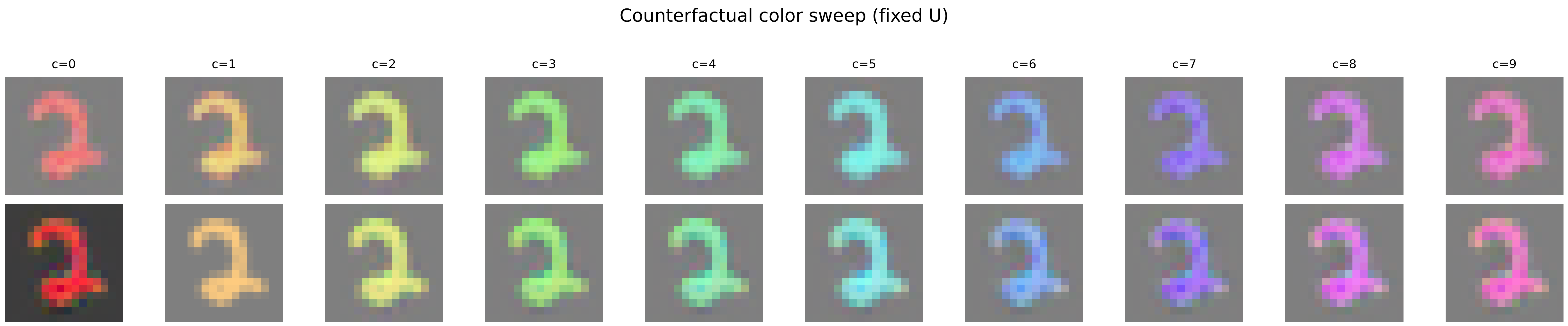
Dataset Summary
Observational tensors:
  LL images:  (1000, 3, 16, 16)  range≈[0.000,1.000]
  LL shapes:  (1000, 1, 16, 16)  range≈[-1.000,1.000]
  digits:     (1000,)
  colors:     (1000,)
  HL obs:     (1000, 21)

Residual artifacts:
  U_ll_hat:   (1000, 3, 16, 16)
  U_hl_hat:   (1000, 1)
```


Residual distributions — sampled pixels
Residuals (pixel 487)
Residuals (pixel 391)
Residuals (pixel 206)
Residuals (pixel 236)
Residuals (pixel 649)
0
50
100
150
200
0.0
0.2
0
20
40
60
80
100
120
140
−0.25
0.00
0.25
0
200
400
600
800
0.0
0.2
0
100
200
300
400
500
600
−0.1
0.0
0.1
0.2
0
20
40
60
80
100
−0.2
0.0
0.2


True image
Pred deterministic
Residual (noise)


Counterfactual color sweep (fixed U)
c=0
c=1
c=2
c=3
c=4
c=5
c=6
c=7
c=8
c=9

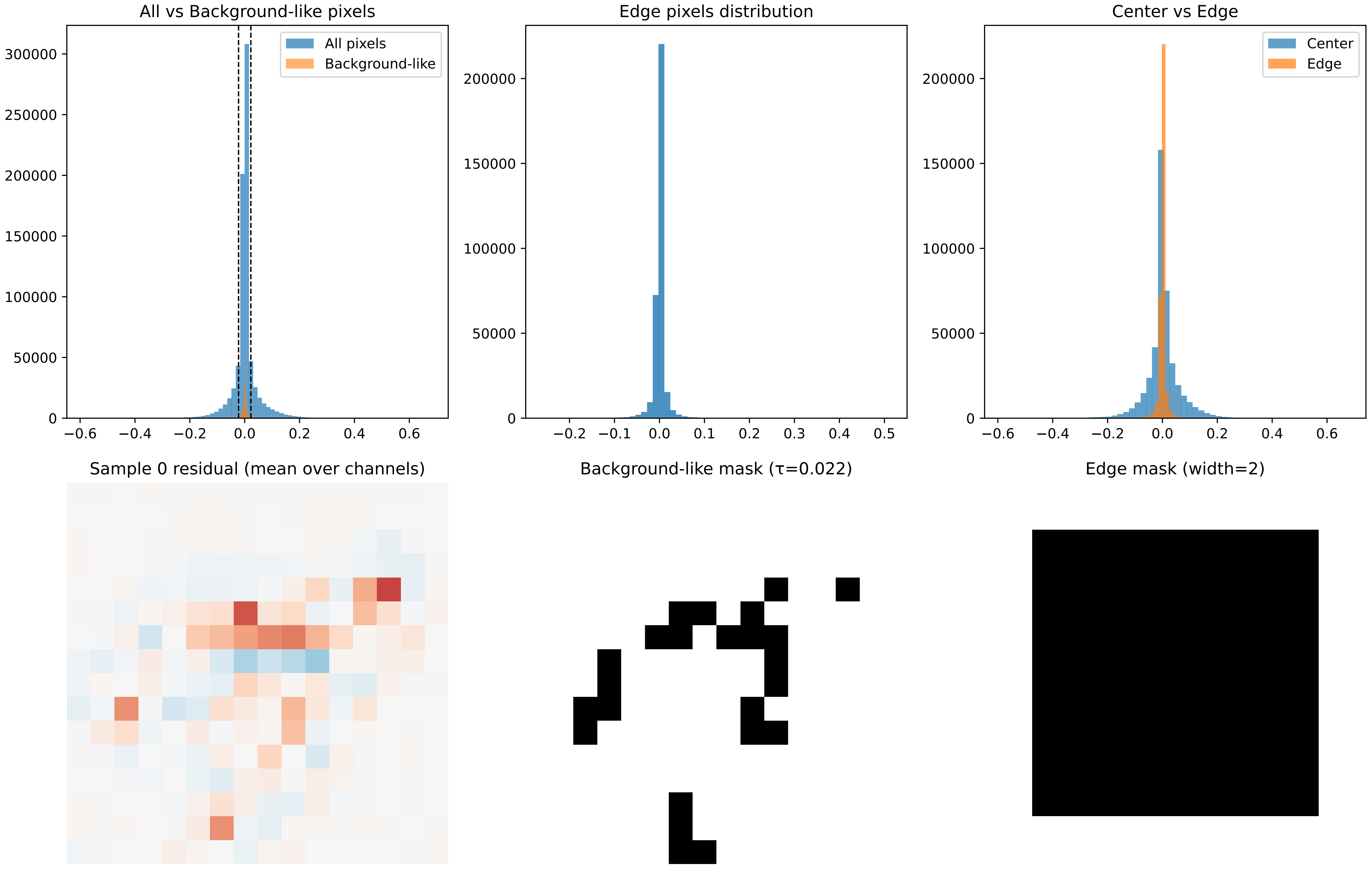
```
Comprehensive Counterfactual Diagnostics — Summary
RESIDUAL DIAGNOSTICS
  R^2(residual ~ [digit,color]) = 0.005296

COUNTERFACTUAL CONSISTENCY (U-Net)
  Digit variance (shape invariance proxy): 0.001315
  Color variance (brightness sensitivity): 0.000289
  Noise stability (mean corr across colors): 1.000000

COLOR HUE DIVERSITY
  Mean pairwise JS divergence (hue histograms): 0.129739
```

```
Residuals — Quick Statistics
U_ll_hat quick stats:
  shape=(1000, 3, 16, 16) dtype=torch.float32
  min=-0.585140 max=0.678585
  mean=0.001990 std=0.044839
  nonzero=768000/768000 (100.00%)

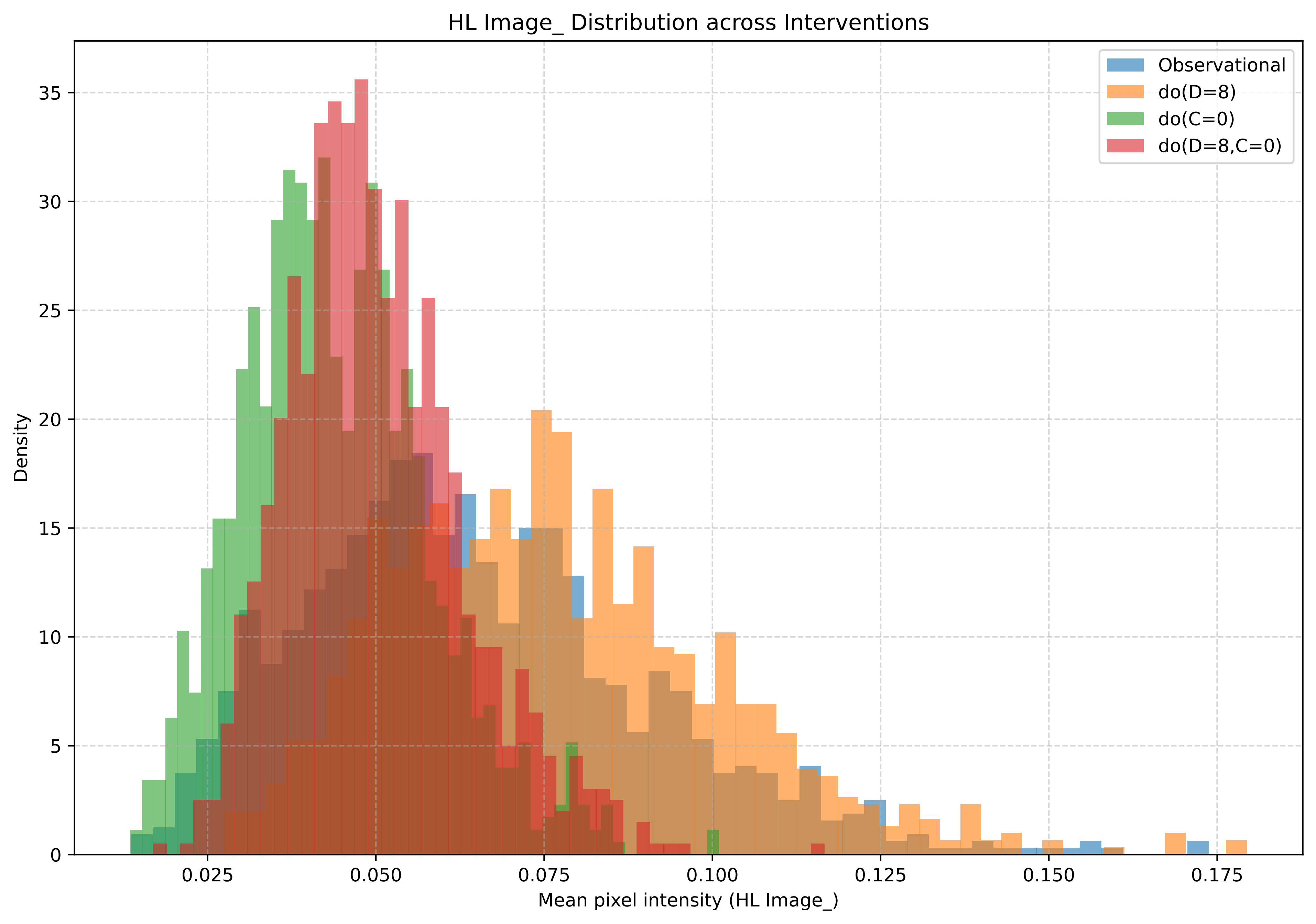
U_hl_hat quick stats:
  shape=(1000, 1) dtype=torch.float32
  min=-0.060246 max=0.072707
  mean=-0.000049 std=0.017795
  nonzero=1000/1000 (100.00%)
```



All vs Background-like pixels
All pixels
Background-like
Edge pixels distribution
Center vs Edge
Center
Edge
Sample 0 residual (mean over channels)
Background-like mask (τ=0.022)
Edge mask (width=2)

```
Background & Edge Analysis (U_ll_hat)
Image size: 16x16 | channels: 3
Background threshold τ: 0.022419
Edge band width: 2 px
Background-like pixels: 554992/768000 (72.26%)
Center stats: mean=0.002616 std=0.057849 min=-0.585140 max=0.678585 count=432000
Edge stats:   mean=0.001186 std=0.017080 min=-0.258781 max=0.511845 count=336000
Near-zero counts:
  exactly 0: 0
  |x| < 1e-3: 86573  |x| < 1e-2: 446427  |x| < 1e-1: 726764
```

```
HL Image_ Distribution – Summary Stats
HL Image_ Distribution across Interventions
Observational:
  Mean: 0.064914 | Std: 0.025783 | Min: 0.013692 | Max: 0.173787
do(D=8):
  Mean: 0.076496 | Std: 0.024477 | Min: 0.027476 | Max: 0.179430
do(C=0):
  Mean: 0.044031 | Std: 0.013803 | Min: 0.013508 | Max: 0.100980
do(D=8,C=0):
  Mean: 0.050018 | Std: 0.012925 | Min: 0.016927 | Max: 0.116666
```



HL Image_ Distribution across Interventions
Observational
do(D=8)
do(C=0)
do(D=8,C=0)
Density
0
5
10
15
20
25
30
35
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Mean pixel intensity (HL Image_)

HL Intervention Response — Mean Shifts

INTERVENTION RESPONSE (mean shift vs obs)

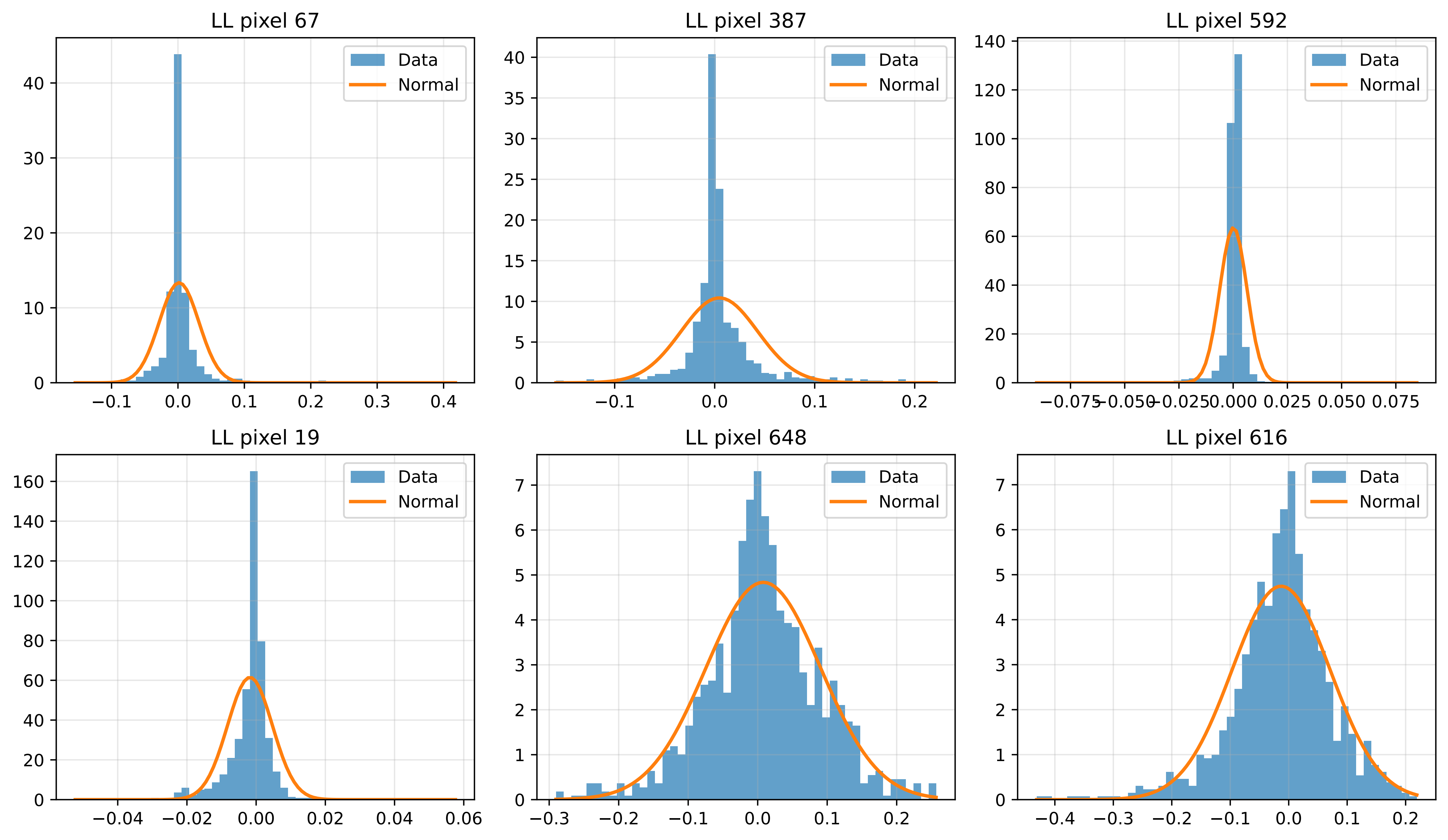
do(D=8): Δ mean = +0.011583

do(C=0): Δ mean = -0.020883

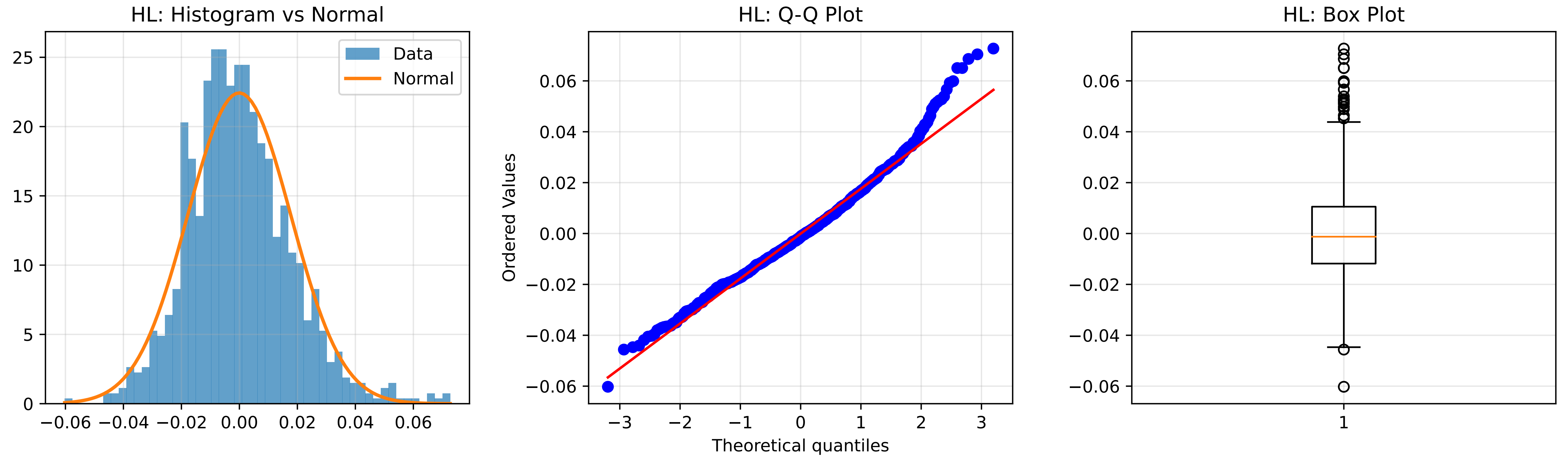
do(D=8,C=0): Δ mean = -0.014895

GAUSSIANITY TESTS FOR NOISE

Gaussianity – LL Stats (first 10)
LOW-LEVEL NOISE (first 10 sampled pixels)
Pixel 67: DP p=0.0000, SW p=0.0000, AD stat=91.9298 (5% crit 0.7840)
Pixel 387: DP p=0.0000, SW p=0.0000, AD stat=74.4692 (5% crit 0.7840)
Pixel 592: DP p=0.0000, SW p=0.0000, AD stat=110.4699 (5% crit 0.7840)
Pixel 19: DP p=0.0000, SW p=0.0000, AD stat=58.8446 (5% crit 0.7840)
Pixel 648: DP p=0.0001, SW p=0.0000, AD stat=3.3266 (5% crit 0.7840)
Pixel 616: DP p=0.0000, SW p=0.0000, AD stat=7.3812 (5% crit 0.7840)
Pixel 476: DP p=0.0000, SW p=0.0000, AD stat=111.7060 (5% crit 0.7840)
Pixel 7: DP p=0.0000, SW p=0.0000, AD stat=24.6107 (5% crit 0.7840)
Pixel 389: DP p=0.0000, SW p=0.0000, AD stat=15.7460 (5% crit 0.7840)
Pixel 334: DP p=0.0000, SW p=0.0000, AD stat=138.9217 (5% crit 0.7840)


LL pixel 67
LL pixel 387
LL pixel 592
LL pixel 19
LL pixel 648
LL pixel 616
Data
Normal

Gaussianity – HL Stats
HIGH-LEVEL NOISE (scalar Image_ residuals)
D'Agostino–Pearson: p=0.0000
Shapiro–Wilk: p=0.0000
Anderson–Darling: stat=3.3140 (5% crit 0.7840)


HL: Histogram vs Normal
Data
Normal
HL: Q-Q Plot
Ordered Values
Theoretical quantiles
HL: Box Plot

Gaussianity – Summary & Interpretation

SUMMARY STATISTICS

LL: mean(mean)=0.001990, mean(std)=0.035820, mean(skew)=0.937319, mean(kurt)=28.998423

HL: mean=-0.000049, std=0.017786, skew=0.538527, kurt=1.061100

INTERPRETATION

LL: ✗ Non-Gaussian indications (mean DP $p \leq 0.05$)

HL: ✗ Non-Gaussian indications (DP $p \leq 0.05$)

Overall Summary & Heuristics

SUMMARY

```
Residual independence (R^2): 0.005296
Digit variance (shape invariance): 0.001315
Color variance (sensitivity):      0.000289
Noise stability (mean corr):       1.000000
Hue diversity (JS):                0.129739
```

Heuristics:

- Residuals ≈ independent (good)
- Shape preserved if digit variance < 1e-2
- Low color sensitivity
- Noise stable if corr > 0.8